Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to CE, EEE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the advantages of modulation.
b) A carrier with amplitude modulated to a depth of 50% by a sinusoidal, produces side band frequencies of 10.005 MHZ and 9.995 MHZ. The amplitude of each side frequency is 20 V. Draw the signal in time domain and frequency domain. [8+7]

2.a) Explain the terms gain, attenuation and why they are generally expressed in decibels.
b) A two stage amplifier has gain of 200 for first stage and 100 for second stage. Express overall gain in dB. [8+7]

3.a) Explain the method of generation and detection of PAM signals with neat schematics.
b) A band limited signal has a bandwidth equal to 4 KHZ. What sampling rate should be used to guarantee a guard band of 1200 Hz? [8+7]

4.a) Draw the constellation diagram for M-ary PSK System.
b) Explain BPSK system transmitter and receiver with the help of diagrams. [7+8]

5.a) Explain the principles of Cross bar switching in Telecommunication.
b) Given that MTBF=2000 Hrs and MTTR=4 Hrs. Calculate the unavailability for single and dual processor systems for 10 years and 5 years. [8+7]

6.a) Explain Routing Control in Packet Switching.
b) What are the differences between LAN, WAN, MAN and Internet? [7+8]

7.a) What are the losses occurred in optical fibers? Explain briefly.
b) Explain the Frequency allocation for satellite services. [7+8]

8.a) Explain the procedure of cell-splitting and advantages of cell sectorization.
b) Describe the RFID communication and its advantages. [7+8]

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